CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-502: Java Programming & Dynamic Webpage Design

Session: December 2021 Examination | **Paper Code:** 18022 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Explain destroy() method, a stage in the life cycle of an Applet.**

Explanation: `destroy()` is the final callback method in the Applet lifecycle, invoked by the browser or AppletViewer before the applet is completely removed from memory and garbage collected. It is called only once after the `stop()` method to release non-memory resources (threads, sockets, database handles).

Q2 (7.5M) **What do you mean by a Package in Java? State the different Access Specifiers available in Java.**

Access Specifiers:

Access Specifier	Same Class	Same Package	Subclass (Diff Pkg)	World (Everywhere)
`private`	Yes	No	No	No
`default` (package-private)	Yes	Yes	No	No
`protected`	Yes	Yes	Yes (via inheritance)	No
`public`	Yes	Yes	Yes	Yes

Q3 (7.5M) **What is Exception? Give different types of exceptions that occur during runtime. Why handle exceptions?**

Explanation: An **Exception** is an abnormal runtime event that disrupts the normal flow of program instructions. Types: Checked (`IOException` , `SQLException`) and Unchecked/Runtime (`ArithmeticException` , `NullPointerException` , `ArrayIndexOutOfBoundsException`). Handled to maintain graceful execution and avoid abrupt program termination.

Q4 (7.5M) **What are the life-cycle methods of a JSP? Differences between JSP Custom Tags and JavaBeans.**

JSP Lifecycle: `jspInit()` → `_jspService(HttpServletRequest, HttpServletResponse)` → `jspDestroy()` .
Differences: JavaBeans encapsulate reusable business data and logic via getter/setter methods. Custom Tags encapsulate presentation logic and HTML formatting within custom XML-style tags.

Q5 (7.5M) **Explain Thread Priority. What are default values? Write a program to define two threads where one prints 1 to 10 and the other prints 10 to 1.**

Thread Priorities: Integer scale from 1 (`MIN_PRIORITY`) to 10 (`MAX_PRIORITY`), with default 5 (`NORM_PRIORITY`).
Program:

```
class ForwardPrinter extends Thread {
    public void run() {
        for(int i = 1; i ⩽ 10; i++) {
            System.out.println("Forward: " + i);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

class BackwardPrinter extends Thread {
    public void run() {
        for(int j = 10; j ⩾ 1; j--) {
            System.out.println("Backward: " + j);
            try { Thread.sleep(200); } catch(Exception e) {}
        }
    }
}

public class PriorityThreadDemo {
    public static void main(String[] args) {
        ForwardPrinter t1 = new ForwardPrinter();
        BackwardPrinter t2 = new BackwardPrinter();
        t1.setPriority(Thread.MAX_PRIORITY);
        t2.setPriority(Thread.MIN_PRIORITY);
        t1.start();
        t2.start();
    }
}
```

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **Explain Client-Server Socket Programming in Java with complete code.**

Socket Programming: `ServerSocket` on server side and `Socket` on client side transmitting data over TCP streams using `DataInputStream` and `DataOutputStream` .

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q7 (22.5M) **Explain JDBC Architecture and write a program to update employee salaries based on department.**

Program:

```
import java.sql.*;

public class UpdateSalaryDemo {
    public static void main(String[] args) {
        try {
            Connection con = DriverManager.getConnection("jdbc:mysql://localhost:3306/db", "root",
            PreparedStatement ps = con.prepareStatement("UPDATE Employee SET Salary = Salary * 1.1
            ps.setString(1, "IT");
            int rows = ps.executeUpdate();
            System.out.println("Salaries updated for " + rows + " employees.");
            con.close();
        } catch(Exception e) { e.printStackTrace(); }
    }
}
```

Q8 (22.5M) **Explain AWT Event Handling with Delegation Event Model. Write a program to handle Button click events.**

Delegation Event Model: An event source generates an event and sends it to registered event listeners implementing listener interfaces (e.g., `ActionListener`).
